

Disproof of Conjecture 00000009898: the torsion lattice of $K(1 \rightarrow 2)$ is the pentagon

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Abstract

Conjecture 00000009898 asserts that the lattice $\text{tors } A$ of torsion classes of a finite-dimensional algebra A is distributive *if and only if* the oriented path partial order of the quiver of A is a forest order, and that the minimal counterexample to distributivity is the three-vertex oriented cycle. Both claims are false. Over *every* field K , the path algebra $A = KA_2$ of the quiver $1 \rightarrow 2$ has a 2-element chain as its path order, which is a forest order under every reading. Yet $\text{tors } A$ is the pentagon N_5 , which is not distributive, and A_2 has only two vertices. The refutation is fully formalised in Lean 4 (core library only, for an arbitrary field). It defines representations, morphisms, quotients, short exact sequences, torsion classes and the lattice operations from scratch. An independent Python script recomputes the whole lattice by brute force over $\mathbb{F}_2$ and $\mathbb{F}_3$. We also prove the correct criterion: $\text{tors } A$ is distributive iff the quiver of A has no arrow between distinct vertices. So the “only if” direction of the conjecture holds, and the “if” direction fails for every quiver with such an arrow.

1 The claim

The conjecture (English version): “*The torsion class lattice of a finite-dimensional algebra is distributive if and only if the oriented path partial order of the quiver is a forest order; the minimal counterexample to distributivity is the three-vertex oriented cycle, and the length of distributive approximations of the lattice is bounded.*”

The statement is a conjunction of three clauses:

- (C1) for every finite-dimensional algebra A : $\text{tors } A$ is distributive $\iff$ the path order of the quiver Q_A is a forest order;
- (C2) the minimal counterexample to distributivity is the oriented 3-cycle. In particular, every algebra whose quiver has fewer than 3 vertices has a distributive torsion lattice;
- (C3) “the length of distributive approximations is bounded”. This is not defined in the statement, and we do not use it.

We refute the direction “ $\Leftarrow$ ” of (C1), hence (C1), and also (C2). One false clause already makes the conjunction false.

2 Set-up

Let K be an arbitrary field and $A = KA_2$ the path algebra of $A_2: 1 \xrightarrow{a} 2$. It has dimension 3, with basis e_1, e_2, a . As usual, finite-dimensional A -modules are the same as finite-dimensional

representations $M = (V_1 \xrightarrow{f} V_2)$: finite-dimensional K -spaces V_1, V_2 with a K -linear map f . A morphism $(g_1, g_2): M \rightarrow N$ is a pair of linear maps with $f_N g_1 = g_2 f_M$. Kernels, images and cokernels are computed vertexwise, so $M \rightarrow N$ is an epimorphism (resp. monomorphism) iff g_1, g_2 are surjective (resp. injective), and a sequence $0 \rightarrow L \rightarrow M \rightarrow N \rightarrow 0$ is exact iff it is exact at both vertices.

The indecomposable representations are

$$S_1 = (K \rightarrow 0), \quad S_2 = (0 \rightarrow K), \quad P_1 = (K \xrightarrow{\text{id}} K).$$

Here S_1, S_2 are the simples, $P_2 = S_2$, and $P_1 = I_2$ is projective-injective.

Definition 1. A *torsion class* is a class $\mathcal{T}$ of modules that contains 0 and is closed under quotients (images of epimorphisms) and extensions. Closure under isomorphisms is a special case of closure under quotients. $\text{tors } A$ is the set of torsion classes ordered by inclusion. It is a complete lattice: the meet is the intersection, and the join $\mathcal{T} \vee \mathcal{U}$ is the intersection of all torsion classes containing $\mathcal{T} \cup \mathcal{U}$.

If one does not require $0 \in \mathcal{T}$, the only new torsion class is $\emptyset$, since a nonempty class closed under quotients contains 0 (as a quotient $M \twoheadrightarrow 0$). All classes used below contain 0, so the argument is unchanged.

The path order. For a quiver Q write $i \leq j$ if there is an oriented path (of length ≥ 0) from i to j . For A_2 this is the chain $1 < 2$. Some authors orient the other way, which gives $2 < 1$, again a chain. A *forest order* is defined in the literature in several ways: every principal down-set is a chain (a rooted forest); every principal up-set is a chain; or the Hasse diagram has no cycle. A 2-element chain satisfies all of these, and it is a partial order. The quiver of $A = KA_2$ (its Gabriel quiver) is A_2 itself, since A is basic with $\text{rad } A / \text{rad}^2 A = Ka$. Hence

the path order of the quiver of KA_2 is a forest order, for every reading.

3 The five torsion classes

Define three classes of representations:

$$x = \{M : V_2 = 0\}, \quad y = \{M : f \text{ surjective}\}, \quad z = \{M : V_1 = 0\}.$$

Also let 0 be the class of zero representations, and $\text{mod } A$ the class of all of them.

Lemma 2. x, y and z are torsion classes.

Proof. All three contain 0.

x . Let $(g_1, g_2): M \twoheadrightarrow N$ with $M \in x$. Then $V_2^N = g_2(V_2^M) = g_2(0) = 0$. Let $0 \rightarrow L \xrightarrow{i} M \xrightarrow{p} N \rightarrow 0$ be exact with $L, N \in x$, and let $v \in V_2^M$. Then $p_2(v) \in V_2^N = 0$, so $v = i_2(u)$ for some $u \in V_2^L = 0$, and $v = 0$.

z . The same argument at vertex 1.

y , *quotients*. Let $w \in V_2^N$. Write $w = g_2(v)$, and $v = f_M(u)$. Then $w = g_2 f_M(u) = f_N(g_1 u)$.

y , *extensions*. Let $w \in V_2^M$. Since $N \in y$, we have $p_2(w) = f_N(a)$ for some a . Since p_1 is onto, $a = p_1(b)$. Then $p_2(w) = f_N p_1(b) = p_2 f_M(b)$, so $w - f_M(b) \in \ker p_2 = \text{im } i_2$, say $w - f_M(b) = i_2(c)$. Since $L \in y$, $c = f_L(d)$, so $i_2(c) = i_2 f_L(d) = f_M(i_1 d)$. Therefore $w = f_M(b + i_1 d)$. $\square$

Lemma 3. *Every torsion class $\mathcal{T}$ with $S_1, S_2 \in \mathcal{T}$ is mod A .*

Proof. The sequences $0 \rightarrow S_i \rightarrow S_i^{n+1} \rightarrow S_i^n \rightarrow 0$ (inclusion of the first coordinate, projection onto the others) are exact. By induction, $S_1^m, S_2^n \in \mathcal{T}$ for all m, n ; the case $n = 0$ is $0 \in \mathcal{T}$. For an arbitrary $M = (K^m \xrightarrow{f} K^n)$ the sequence

$$0 \rightarrow (0 \rightarrow K^n) \xrightarrow{(0, \text{id})} M \xrightarrow{(\text{id}, 0)} (K^m \rightarrow 0) \rightarrow 0$$

is exact. Its outer terms are S_2^n and S_1^m , so $M \in \mathcal{T}$. □

Proposition 4. $\text{tors}(KA_2) = \{0, x, y, z, \text{mod } A\}$. *These classes form the pentagon N_5 :*

$$0 \subsetneq x \subsetneq y \subsetneq \text{mod } A, \quad 0 \subsetneq z \subsetneq \text{mod } A,$$

with z incomparable to x and to y . Also $x = \text{add } S_1$, $y = \text{add}\{S_1, P_1\}$ and $z = \text{add } S_2$.

Proof. If $f: V_1 \rightarrow V_2$ has rank r , choose a complement C of $\ker f$ and a complement D of $\text{im } f$. In adapted bases, $M \cong P_1^r \oplus S_1^{\dim V_1 - r} \oplus S_2^{\dim V_2 - r}$. A torsion class is closed under direct sums (split extensions) and under direct summands (quotients). So it is add of the set of indecomposables it contains. That set $I \subseteq \{S_1, S_2, P_1\}$ satisfies $P_1 \in I \Rightarrow S_1 \in I$, because $P_1 \twoheadrightarrow S_1$. It also satisfies $S_1, S_2 \in I \Rightarrow P_1 \in I$, by Lemma 3 or by the extension $0 \rightarrow S_2 \rightarrow P_1 \rightarrow S_1 \rightarrow 0$. This leaves the possibilities $\emptyset$, $\{S_1\}$, $\{S_2\}$, $\{S_1, P_1\}$ and $\{S_1, S_2, P_1\}$. They give 0 , x , z , y and $\text{mod } A$, which are torsion classes by Lemma 2. The inclusions are witnessed by $S_1 \in x \setminus 0$, $P_1 \in y \setminus x$, $S_2 \in z \setminus y$ and $S_1 \in x \setminus z$. □

Only Lemmas 2 and 3 are needed for the disproof. The classification in Proposition 4 shows that the counterexample is the whole lattice, not just a sublattice.

4 Failure of distributivity

Theorem 5. *For every field K , $\text{tors}(KA_2)$ is not distributive:*

$$y \wedge (x \vee z) = y \neq x = (y \wedge x) \vee (y \wedge z).$$

The dual law fails as well: $x \vee (y \wedge z) = x \neq y = (x \vee y) \wedge (x \vee z)$.

Proof. By Lemma 3, $x \vee z = \text{mod } A$, because $S_1 \in x$ and $S_2 \in z$. So $y \wedge (x \vee z) = y$. Next, $y \wedge x = x$ since $x \subseteq y$ (if $V_2 = 0$, f is onto). Also $y \wedge z = 0$: if $V_1 = 0$ and f is onto, then $V_2 = f(0) = 0$. Since x is a torsion class containing $x \cup 0$, we get $(y \wedge x) \vee (y \wedge z) = x$. Finally $P_1 \in y \setminus x$. For the dual law, $x \vee y = y$ and $x \vee 0 = x$. □

Corollary 6. *Clause (C1) is false: for $A = KA_2$ the path order is a forest order but $\text{tors } A$ is not distributive. Clause (C2) is false: A_2 has $2 < 3$ vertices and a non-distributive torsion lattice. Since a one-vertex algebra A is local, with $\text{tors } A = \{0, \text{mod } A\}$ (a 2-chain), the true minimum number of vertices is 2.*

5 Readings covered

- **Field.** Everything holds over every field. The Lean proof is for an arbitrary field, with instances $\mathbb{F}_2$ and $\mathbb{F}_3$.
- **Forest order.** The path order of A_2 is a 2-chain: a partial order whose down-sets and up-sets are chains and whose Hasse diagram is a single edge. It is a forest under every definition and for either orientation convention (left or right modules, i.e. $1 < 2$ or $2 < 1$). Lean uses the strongest version (antisymmetric, and both down-sets and up-sets are chains).
- **Distributivity.** Both distributive laws fail, so any definition of a distributive lattice is covered.
- **Torsion classes.** The result is the same whether or not $\emptyset$ is admitted, and closure under isomorphism is automatic.
- **Minimality (C2).** Measured by vertices (or arrows), A_2 is smaller than the 3-cycle. If “minimal counterexample” means a minimal quiver whose path order is *not* a forest, the 2-cycle $1 \rightrightarrows 2$ is smaller still. With relations $\text{rad}^2 = 0$ this is the preprojective algebra of A_2 , whose torsion lattice is the weak order of S_3 , a hexagon.
- **(C3)** is undefined. The disproof does not depend on it.

6 Remark: the correct criterion

The following is not needed for the disproof. It shows that only the “if” half of (C1) fails.

Proposition 7. *Let A be a finite-dimensional basic algebra over a field. Then $\text{tors } A$ is distributive if and only if the quiver of A has no arrow between two distinct vertices. In that case the path order is an antichain, hence a forest order.*

Proof. Suppose there is an arrow between distinct vertices $i \neq j$. Then $\text{Ext}^1(S_i, S_j) \neq 0$, possibly after exchanging i and j according to the convention, so there is a non-split extension $0 \rightarrow S_j \rightarrow M \rightarrow S_i \rightarrow 0$. For a class $\mathcal{C}$ let $T(\mathcal{C})$ be the smallest torsion class containing it. The class $\{X : \text{Hom}(X, S_j) = 0\}$ is closed under quotients and extensions (left exactness of $\text{Hom}(-, S_j)$), and it contains M : a nonzero map $M \rightarrow S_j$ would split the sequence. Hence $S_j \notin T(M)$, and since every nonzero $X \in T(S_j) \subseteq \text{Filt}(S_j)$ has $\text{Hom}(X, S_j) \neq 0$, we get $T(M) \wedge T(S_j) = 0$. Moreover $S_i \in T(M)$, and $M \in T(S_i) \vee T(S_j)$. But $M \notin T(S_i) \subseteq \text{Filt}(S_i)$. Thus $T(M) \wedge (T(S_i) \vee T(S_j)) = T(M) \neq T(S_i) = (T(M) \wedge T(S_i)) \vee (T(M) \wedge T(S_j))$.

Conversely, if no arrow joins distinct vertices, then $\text{Ext}^1(S_i, S_j) = 0$ for $i \neq j$. The blocks of A correspond to the connected components of its quiver, so $A = \prod_i A_i$ with each A_i local. A nonzero torsion class of a local algebra contains the unique simple module, hence all modules. So $\text{tors } A \cong \prod_i \{0 < 1\}$ is Boolean, hence distributive. $\square$

So the “only if” direction of (C1) is true. The “if” direction fails for every quiver with an arrow between distinct vertices, and A_2 is the smallest example.

7 Machine verification

Lean 4 (`lean4/Main.lean`, **core only**, **v4.19.0**). Everything is defined from scratch:

- a field structure `Fld` (all field axioms), coordinate spaces K^n and linear maps;
- representations `Rep K` of A_2 (dimensions d_1, d_2 arbitrary, f any linear map), morphisms `Hom`, epimorphisms, isomorphisms, short exact sequences;
- `IsTors` (contains 0, closed under quotients and extensions), meets and joins, with proofs that they are torsion classes and are the greatest lower and least upper bounds;
- the quiver A_2 , oriented paths, `ForestOrder`.

Main results: `xC_isTors`, `yC_isTors`, `zC_isTors` (Lemma 2, for all dimensions); `all_of_S1_S2` (Lemma 3); `pentagon` (the N_5 relations); `not_distributive`, `not_codistributive` (Theorem 5); `a2_forest`; and

`conjecture_00000009898_false (K) [Fld K] : ¬ IfClause K,`

where `IfClause K := ForestOrder A2Q → Distributive K`. Also `iffClause_false` and `minClause_false`.

Non-vacuity: `five_tors` shows that $0, x, y, z, \text{mod } A$ are torsion classes and that $\{f \text{ injective}\} = \text{add}\{S_2, P_1\}$ is not. Field instances are provided for $\mathbb{F}_2$ and $\mathbb{F}_3$. `c3_facts` records that the 3-cycle has 3 vertices and a non-antisymmetric path relation. The axioms used are only `propext`, `Quot.sound` and `Classical.choice`.

Limitations. Representations use coordinate spaces K^d . Every finite-dimensional space is isomorphic to one, and torsion classes are closed under isomorphism, so this loses nothing. The equivalence between KA_2 -modules and representations of A_2 , the classification in Proposition 4, and the general criterion are proved only in this report.

Python (`verify.py`, **standard library**). A different method: over $\mathbb{F}_2$ and $\mathbb{F}_3$ it enumerates all representations with $d_1, d_2 \leq 2$ and computes isomorphism classes by brute force under $GL \times GL$ (14 classes). It finds the quotient relation by searching for surjective morphisms, and the extension triples from kernels of surjective morphisms. It then tests all 2^{14} subsets for being torsion classes. It finds exactly five, equal to $0, x, y, z, \text{mod } A$, forming N_5 . Distributivity fails, and exhaustively 2 of the 125 triples violate it. It also checks the path orders of A_2 and of the 3-cycle.